Question

The reaction of metallic potassium with carbon monoxide can produce a black powder with the chemical composition KCO.

It is known that the powder composition includes two substances, $\mathbf{A}$ and $\mathbf{B}$. $\mathbf{A}$ reacts with dilute hydrochloric acid to produce a monocarboxylic acid. $\mathbf{B}$ reacts with dilute hydrochloric acid to produce a substance $\mathbf{D}$ containing only one type of functional group, a hydroxyl group. $\mathbf{D}$ is oxidized by chlorine gas in aqueous solution to produce a substance $\mathbf{E}$ with 2 waters of crystallization. $\mathbf{E}$ produces $\mathbf{F}$ under high vacuum heating, where all atoms have the same chemical environment and are extremely unstable.

Which of the following statements is correct?

A. All other options are incorrect

B. $\mathbf{A}$ only has a two-fold rotation axis

C. The anion of $\mathbf{B}$ does not possess strong reducing properties.

D. $\mathbf{D}, \mathbf{E}, \mathbf{F}$ are all ternary compounds

E. Chemical equation for the synthesis of $\mathbf{E}$, when the coefficients are balanced to the simplest integer ratio, the sum of the reactant coefficients and the sum of the product coefficients have a prime number.

Answer

Correct Answer: E

Detailed Explanation

The anions of $\mathbf{A, B}$ obviously have the simplest chemical formula CO^-. Since $\mathbf{A}$ reacts with acid to produce a monocarboxylic acid, the anion of $\mathbf{A}$ can only be $C_2O_2^{2-}$, with the structure [O-]C#C[O-], which hydrolyzes to produce $HOCH_2COOH$. $C_2O_2^{2-}$ has a C_∞ axis, so option B is incorrect.

CHECKPOINT 1 PTS

The anion of $\mathbf{A}$ can only be $C_2O_2^{2-}$, with the structure [O-]C#C[O-]

CHECKPOINT 1 PTS

$C_2O_2^{2-}$ has a C_∞ axis

$\mathbf{D}$, produced by the reaction of $\mathbf{B}$ with acid, has only one type of functional group, the hydroxyl group, which is obviously hexahydroxybenzene, with the structure OC1=C(O)C(O)=C(O)C(O)=C1O, and the chemical formula $C_6(OH)_6$. Therefore, the structure of the anion of $\mathbf{B}$ is [O-]C1=C([O-])C([O-])=C([O-])C([O-])=C1[O-], which has strong reducing properties, so option C is incorrect.

CHECKPOINT 1 PTS

The structure of **D** is OC1=C(O)C(O)=C(O)C(O)=C1O

CHECKPOINT 1 PTS

The structure of the anion of **B** is [O-]C1=C([O-])C([O-])=C([O-])C([O-])=C1[O-]

CHECKPOINT 1 PTS

The anion of **B** has strong reducing properties

D is oxidized in aqueous solution, and should produce cyclohexanhexone, but it is extremely unstable. The carbonyl groups are extremely electron-deficient and are easily attacked by water nucleophilically to form carbonyl hydrates. Therefore, the chemical formula of **E** is $C_6(OH)_{12} \cdot 2H_2O$, then obviously **F** is cyclohexanhexone, with the structure O=C(C(C(C(C1=O)=O)=O)=O)C1=O.

CHECKPOINT 1 PTS

The chemical formula of **E** is $C_6(OH)_{12} \cdot 2H_2O$

CHECKPOINT 1 PTS

F is cyclohexanhexone, with the structure O=C(C(C(C(C1=O)=O)=O)=O)C1=O

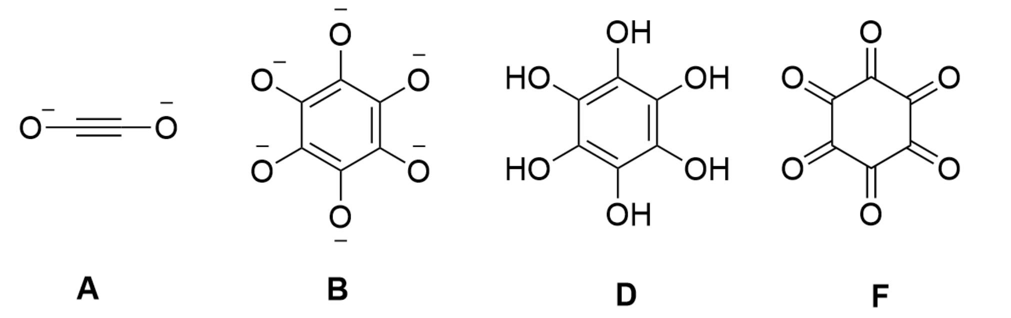
The chemical equation for the formation of **E** is:

$$C_6(OH)_6 + 3Cl_2 + 8H_2O = C_6(OH)_{12} \cdot 2H_2O + 6HCl$$

CHECKPOINT 1 PTS

$$C_6(OH)_6 + 3Cl_2 + 8H_2O = C_6(OH)_{12} \cdot 2H_2O + 6HCl$$

According to the chemical formulas and equation, option E is correct, and D is incorrect.



The structure of the anion of **A** is [O-]C#C[O-]; the structure of **D** is OC1=C(O)C(O)=C(O)C(O)=C1O; the structure of **F** is O=C(C(C(C(C1=O)=O)=O)=O)C1=O; the structure of the anion of **B** is [O-]C1=C([O-])C([O-])=C([O-])C([O-])=C1[O-]